

Code:

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% Modeling Random Direction
%% Define parameters
stepsize = 1;
num_trials = 100; % Number of trials
% Initialize variables
average_ratio_1000 = 0;
average_ratio_5000 = 0;
ratio_of_ratios = 0;
iteration = 0;
while abs(ratio_of_ratios - 5) > 0.01 % Set a tolerance of 0.01 for
convergence
    % Reset averages and ratios for each iteration
    ratios_1000 = zeros(num_trials, 1);
    ratios_5000 = zeros(num_trials, 1);

    for trial = 1:num_trials
        % Start with initial position of molecules
        x1 = 0;
        y1 = 0;
        x2 = 0;
        y2 = 0;
        % Generate random steps for 1000 steps
        numsteps1 = 1000;
        x1_coords = zeros(numsteps1, 1);
        y1_coords = zeros(numsteps1, 1);
        for i = 1:numsteps1
            angle1 = 2*pi*rand;
            dx1 = stepsize*cos(angle1);
            dy1 = stepsize*sin(angle1);
            x1 = x1 + dx1;
            y1 = y1 + dy1;
            x1_coords(i) = x1;
            y1_coords(i) = y1;
        end
        % Generate random steps for 5000 steps
        numsteps2 = 5000;
        x2_coords = zeros(numsteps2, 1);
        y2_coords = zeros(numsteps2, 1);
        for i = 1:numsteps2
            angle2 = 2*pi*rand;
            dx2 = stepsize*cos(angle2);
            dy2 = stepsize*sin(angle2);
            x2 = x2 + dx2;
            y2 = y2 + dy2;
            x2_coords(i) = x2;
            y2_coords(i) = y2;
        end
        % Calculate ratio for this trial
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        ratio_1000 = norm([x1_coords(end), y1_coords(end)])^2;
        ratio_5000 = norm([x2_coords(end), y2_coords(end)])^2;
        % Store ratios for this trial
        ratios_1000(trial) = ratio_1000;
        ratios_5000(trial) = ratio_5000;
    end
    % Calculate the average ratios over all trials
    average_ratio_1000 = mean(ratios_1000);
    average_ratio_5000 = mean(ratios_5000);
    % Calculate the ratio of ratios
    ratio_of_ratios = average_ratio_5000 / average_ratio_1000;

    % Increment iteration counter
    iteration = iteration + 1;
end
% Display results
disp(['Ratio of average ratios: ', num2str(ratio_of_ratios)]);
% Plot paths for the last trial
figure;
subplot(1, 2, 1);
plot(x1_coords, y1_coords);
title('Path for 1000 Steps');
xlabel('X-coordinate');
ylabel('Y-coordinate');
axis equal;
subplot(1, 2, 2);
plot(x2_coords, y2_coords);
title('Path for 5000 Steps');
xlabel('X-coordinate');
ylabel('Y-coordinate');
axis equal;

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Short Write Out:

To obtain a ratio closer to 5, I created a loop that made many trials of 5000 random steps and 1000 random steps, averaged them out, and kept iterating them until I got a ratio of 5 with a tolerance of 0.01. This was a success as my program gave me nice plots and very close to the desired ratio.